Neuromorphic ASR

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- Similar to the Interspeech 2025 paper. https://arxiv.org/pdf/2505.24721
- Curiousity about how a transducer model, more involved than CTC, would behave on memristor.
- Study the impacts of offloading various components of the transducer model to device.
- Study the actual impact of noise
- Performance impact on Lexfree vs Tree searches.
- Argue from a LM POV, impact of LM on CTC-RnnT

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Some Prerequisites

- Memristor programming (programming table from paper?) and non idealities
- Memristor crossbars and bitlines to represent model weights and thus the need of QAT.
- Figure of the model topologies and the components, and the components that can be deployed on device.

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- Our matrix is 3 (model types) x 5 (search types) x 3 'stages' (baseline, qat, memristor)
- Primary tables could be 1 for each model type.
- Appendix - Tables across models, but same stages for easy comparison.
- Device deployment changes the interplay between prior/ILM (higher is better) and LM for performance.

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- Error analysis?
- denoising (using mlp or something else), search params, LM params, softmax temperature.
- ft of model
- comparing across different datasets (Data condition)
- Effect of noise introduction during model training

Reference